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OPERATING SYSTEMS LABORATORY

Course Code: CSE309R01

Semester: V

Lab Manual

2026

ROHITH KANNA GV

128003214

A-44

**SCHOOL OF COMPUTING
SHANMUGHA ARTS, SCIENCE, TECHNOLOGY AND RESEARCH
ACADEMY
(SASTRA Deemed to be University)
Tirumalaisamudram, Thanjavur-613 401**

Exercise No. 2 IPC Using Shared Memory System Call

Write program to carry out inter-process communication using shared memory Objectives

To implement IPC using shared memory concept with the help of the library functions available.

Prerequisite

- Knowledge of IPC, Shared memory functions, their syntaxes and

functionalities **Shared Memory:**

- Shared memory is a memory shared between two or more processes. Each process has its own address space;

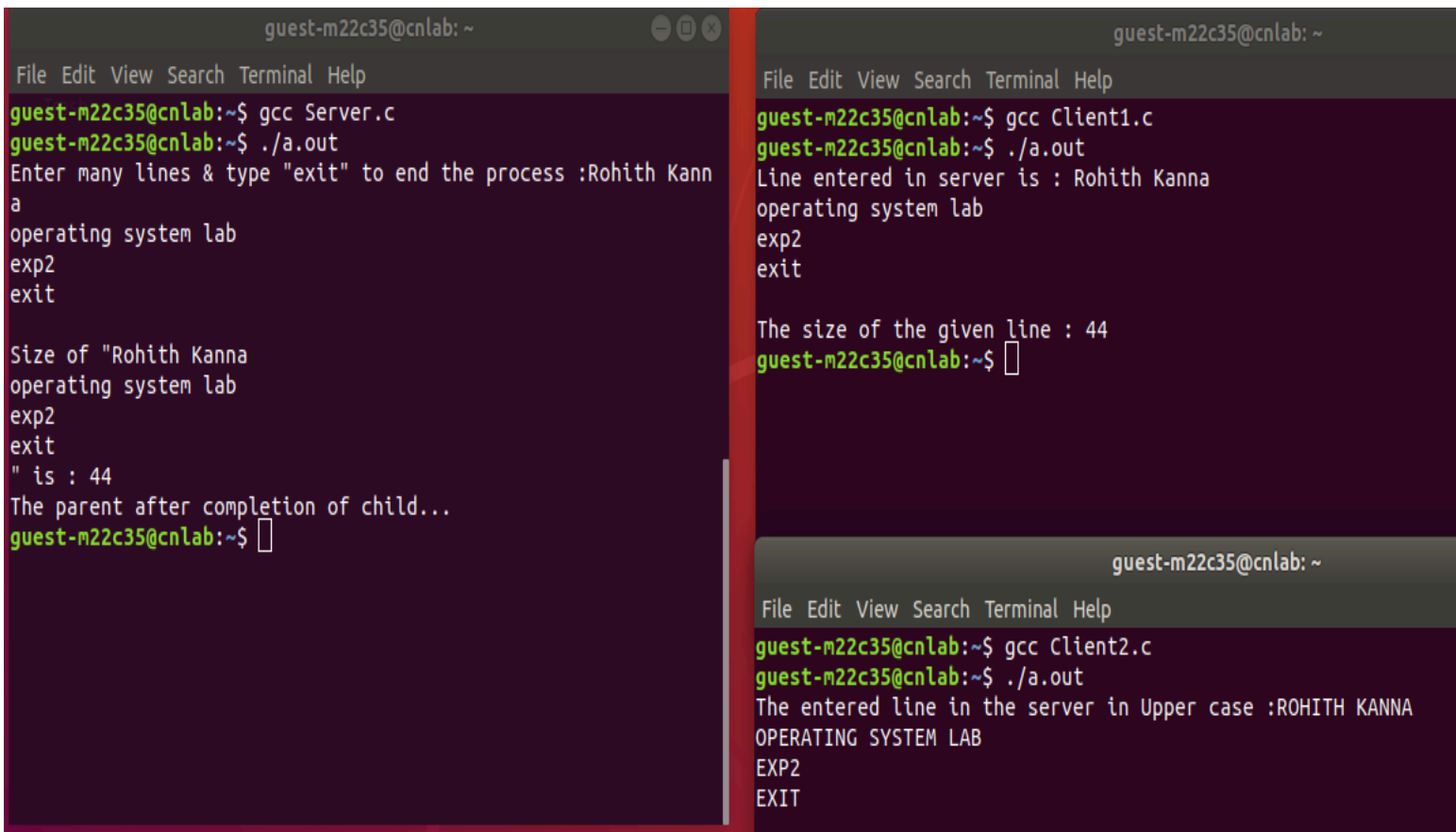
Procedure

- ☐ Create the sender process and receiver process
- ☐ Create a shared memory making using the appropriate function
- ☐ Sender pushes its message into shared memory
- ☐ Receiver retrieves the message and displays it to the user

Sample Input:

Rohith Kanna
operating system lab
exp2
exit

Output:



```
guest-m22c35@cnlab: ~  
File Edit View Search Terminal Help  
guest-m22c35@cnlab:~$ gcc Server.c  
guest-m22c35@cnlab:~$ ./a.out  
Enter many lines & type "exit" to end the process :Rohith Kanna  
a  
operating system lab  
exp2  
exit  
  
Size of "Rohith Kanna  
operating system lab  
exp2  
exit  
" is : 44  
The parent after completion of child...  
guest-m22c35@cnlab:~$
```

```
guest-m22c35@cnlab: ~  
File Edit View Search Terminal Help  
guest-m22c35@cnlab:~$ gcc Client1.c  
guest-m22c35@cnlab:~$ ./a.out  
Line entered in server is : Rohith Kanna  
operating system lab  
exp2  
exit  
  
The size of the given line : 44  
guest-m22c35@cnlab:~$
```

```
guest-m22c35@cnlab: ~  
File Edit View Search Terminal Help  
guest-m22c35@cnlab:~$ gcc Client2.c  
guest-m22c35@cnlab:~$ ./a.out  
The entered line in the server in Upper case :ROHITH KANNA  
OPERATING SYSTEM LAB  
EXP2  
EXIT
```

Observation

Summary of Shared Memory System Calls: Write the syntax of the following in detail:

1. shmget(): Creates or accesses a shared memory segment.

Syntax: int shmget(key_t key, size_t size, int shmflg);

Purpose: Creates a new shared memory segment or accesses an existing

shared memory segment.

2. shmat(): Attaches the shared memory segment to the calling process's address space.

Syntax: void *shmat(int shmid, const void *shmaddr, int shmflg);

Purpose: Attaches the shared memory segment to the address space of the calling process.

3. shmdt(): Detaches the shared memory segment from the process's address space.

Syntax: int shmdt(const void *shmaddr);

Purpose: Detaches the shared memory segment from the process's address space.

4. shmctl(): Controls and manages shared memory segments (e.g., removing, setting permissions, retrieving status).

Syntax: int shmctl(int shmid, int cmd, struct shmid_ds *buf);

Purpose: Controls the shared memory segment, such as removing it, changing permissions, or obtaining its status.

Result

Thus, the program to implement Inter-Process Communication (IPC) using Shared Memory was executed successfully. The sender process stored the data in shared memory, and the receiver process accessed and displayed the data successfully.